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KGiSL Campus, Saravanampatti, Coimbatore – 641 035

Credit Card Billing and Payment Reminder System

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SUBJECT : JAVA PROGRAMMING

Introduction

The Credit Card Billing and Payment Reminder System is a software application designed to help users manage their credit card bills efficiently. It tracks billing dates, due dates, and outstanding balances, and sends timely reminders to avoid late payments. This system reduces the risk of penalties, improves financial discipline, and ensures smooth credit management.

PROBLEM STATEMENT:

1. Late payment fees
2. Increased interest charges
3. Poor credit scores
4. Financial mismanagement.

OBJECTIVES OF THE PROJECT

1. To store and manage credit card details securely
2. To track billing cycles and due dates
3. To send timely reminders for bill payments
4. To avoid late payment penalties
5. To provide a simple and user-friendly interface
6. To improve financial awareness and discipline

SCOPE OF THE SYSTEM

1. Used by individual credit card users
2. Tracks multiple credit cards
3. Sends alerts via system notification
4. Stores transaction and billing details

Can be extended to mobile apps or web-based systems

SYSTEM MODULES

Account & Billing Module– Manages customer/card data, generates statements, and calculates dues with interest.

Payment & Notification Module– Processes payments and sends reminders/alerts for due and overdue bills.

SYSTEM WORKFLOW

Transactions are recorded and compiled during the billing cycle.

System generates the credit card statement with total due and due date.

Reminders are sent to customers before and on the due date. Payment is processed if delayed, late fees are applied and alerts are sent.

JAVA CONCEPTS USED

Classes and Objects (User, Card, Transaction)

Encapsulation (private data with getters/setters)

Inheritance (optional for extending features)

Collections (ArrayList for storing data)

Exception Handling (invalid inputs)

File Handling / Database Connectivity (data storage)

Date and Time API (for reminders)

JAVA PROGRAM

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Main.java

```java
import java.util.*;

class CreditCard {
    private String customerName;
    private double balance;
    private double creditLimit;
    private double dueAmount;

    CreditCard(String name, double limit) {
        this.customerName = name;
        this.creditLimit = limit;
        this.balance = 0;
        this.dueAmount = 0;
    }

    void addTransaction(double amount) {
        if (balance + amount <= creditLimit) {
            balance += amount;
            System.out.println("Transaction added: " + amount);
        } else {
            System.out.println("Credit limit exceeded!");
        }
    }

    void generateBill() {
        dueAmount = balance + (balance * 0.02); // 2% interest
        System.out.println("\nBill Generated for " + customerName);
        System.out.println("Outstanding Balance: " + balance);
        System.out.println("Total Due (with interest): " + dueAmount);
    }

    void sendReminder() {
        if (dueAmount > 0) {
            System.out.println("Reminder: Dear " + customerName + ", your due amount is " + dueAmo
        } else {
```

input

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Main.java

```java
        if (dueAmount > 0) {
            System.out.println("Reminder: Dear " + customerName + ", your due amount is " + dueAmo
        } else {
            System.out.println("Reminder: Dear " + customerName + ", you have no dues. Thank you!"
        }
    }

    void makePayment(double amount) {
        if (amount > 0 && amount <= dueAmount) {
            dueAmount -= amount;
            balance -= amount;
            if (balance < 0) balance = 0;
            System.out.println("Payment successful: " + amount);
        } else {
            System.out.println("Invalid payment amount!");
        }
    }
}

public class Main {   // ✅ Matches file name Main.java
    public static void main(String[] args) {
        CreditCard card = new CreditCard("John Doe", 50000);

        card.addTransaction(10000);
        card.addTransaction(5000);

        card.generateBill();
        card.sendReminder();

        card.makePayment(8000);

        card.sendReminder();
    }
}
```

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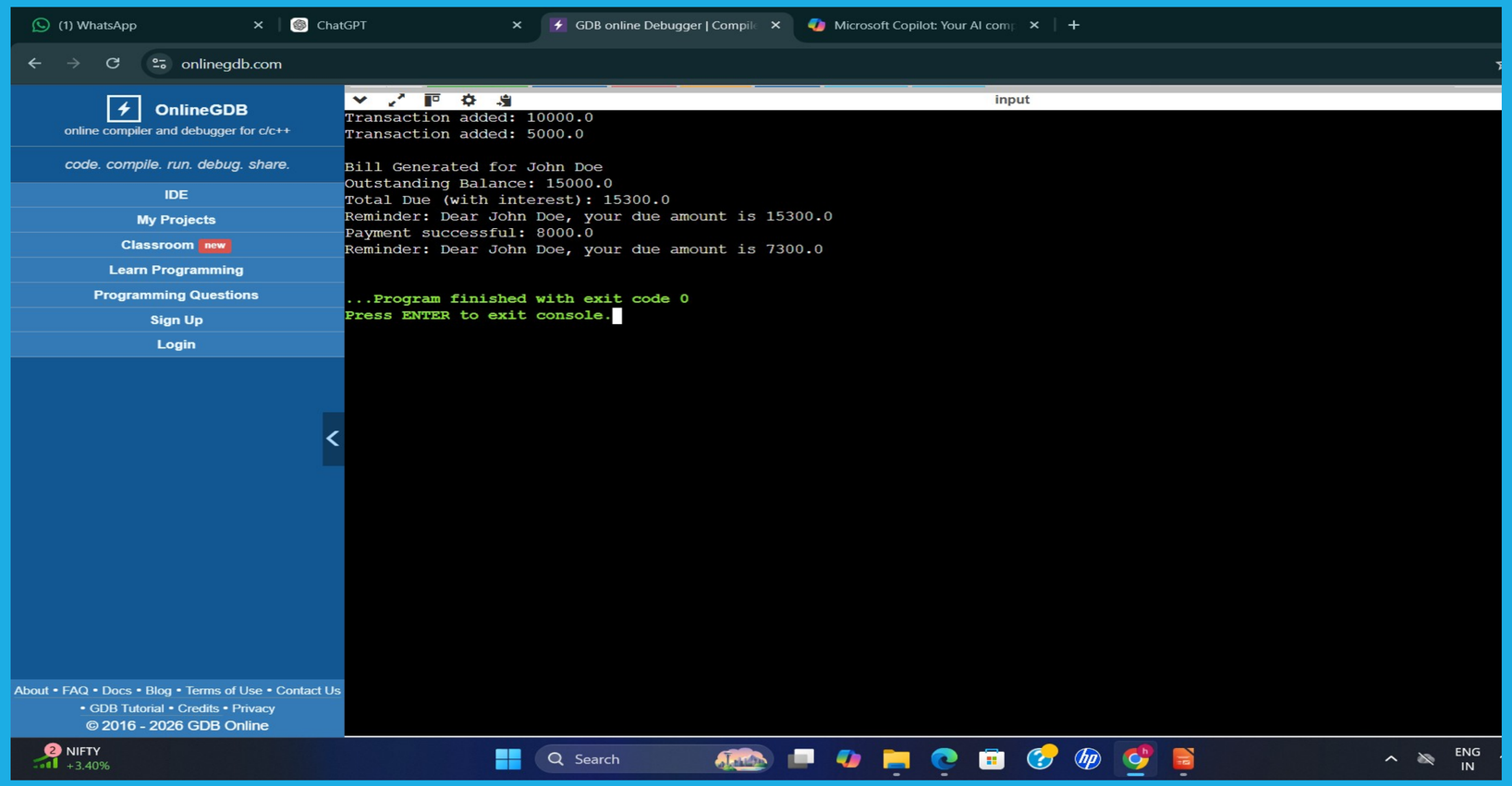
OUTPUT SCREEN



DEMONSTRATION STEPS

1. Create a credit card account and set the credit limit.

2. Add transactions to update the balance during the billing cycle.

3. Generate the bill and send payment reminders to the customer.

4. Process payment and update the remaining due or apply late fees if unpaid.

CONCLUSION

The Credit Card Billing and Payment Reminder System streamlines the management of credit card accounts by automating transaction tracking, bill generation, and payment reminders. It helps customers stay informed about their dues, reduces late payments, and improves financial discipline. For banks and financial institutions, it ensures accurate billing, timely collection, and enhanced customer satisfaction through efficient notifications. Overall, it enhances convenience, transparency, and reliability in credit card management.

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Herbert Schildt, Java: The Complete Reference, McGraw-Hill, 2021 – Covers Java programming concepts applicable to billing systems.

Budi Kurniawan, Java 8 for Programmers, Pustaka Digital, 2019 – Demonstrates practical Java applications including transaction and payment processing.